

1 Solution — GIAM 2.7

Problem 6

1.1 Problem

Sometimes in constructing a proof we find it necessary to “weaken” an inequality. For example, we might already have deduced that $x < y$ but what we need is $x \leq y$. It is okay to deduce $x \leq y$ from $x < y$ because the former is just shorthand for $(x < y) \vee (x = y)$. What rule of inference are we using?

1.2 Answer

Disjunctive addition — from P infer $P \vee Q$.

Take P to be " $x < y$ " and Q to be " $x = y$ ". Since $x \leq y$ is *by definition* the disjunction $(x < y) \vee (x = y)$, the step is

$$\frac{x < y}{\therefore (x < y) \vee (x = y), \quad \text{i.e.} \quad x \leq y.}$$

1.3 Why This Is the Rule

Once $\leq$ is unpacked into its definition, the "weakening" is a single application of a familiar rule.

by definition: $x \leq y$ is $(x < y) \vee (x = y)$
" $\leq$ " is not a primitive symbol — it abbreviates a disjunction

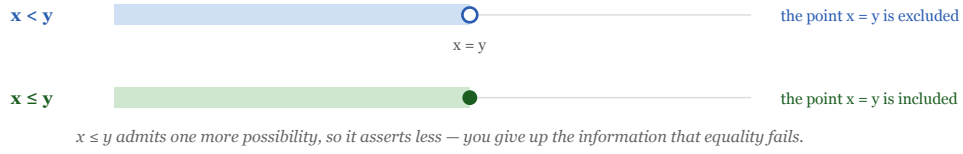
So the step in question is:

$$\frac{x < y}{\therefore (x < y) \vee (x = y)}$$

that is, $x \leq y$

DISJUNCTIVE ADDITION

Why it is a weakening: the claim covers strictly more cases



Going back the other way needs a different rule

weaken:	$x < y \vdash x \leq y$	<i>disjunctive addition</i>
restore:	$x \leq y, x \neq y \vdash x < y$	<i>disjunctive syllogism</i>

Figure 1.1: Top: by definition, x less-than-or-equal-to y is the disjunction x less than y , or x equals y — the symbol is not primitive but abbreviates a disjunction. The step in question is therefore: from x less than y , infer the disjunction x less than y or x equals y , that is x less-than-or-equal-to y — labelled DISJUNCTIVE ADDITION. Below, a number-line picture showing why this is a weakening. The band for x less than y stops at an open circle at the point x equals y , which is excluded. The band for x less-than-or-equal-to y is identical but ends in a filled circle, including the point x equals y . So the weaker claim admits one more possibility and asserts less — you give up the information that equality fails. A final panel contrasts the two directions: to weaken, from x less than y infer x less-than-or-equal-to y , by disjunctive addition; to restore, from x less-than-or-equal-to y together with x not equal to y infer x less than y , by disjunctive syllogism.

The whole exercise turns on recognising that $\leq$ **is not a primitive symbol**. Once it is unpacked into $(x < y) \vee (x = y)$, the step is transparently an instance of

$$P \vdash P \vee Q,$$

which is valid because a disjunction is true as soon as *one* disjunct is. The rule was verified by truth table, and the concrete claim " $x < y$ implies $x \leq y$ " on 2000 random integer pairs.

1.4 Remark — “Weakening” Is Exactly the Right Word

The problem’s scare quotes around *weaken* are well earned: the new statement really does say **less**. Asserting $x < y$ rules out two of the three possibilities allowed by trichotomy ($x = y$ and $x > y$); asserting $x \leq y$ rules out only one. You have thrown away the information that equality fails.

That is why disjunctive addition can never lead you astray — it only ever *discards* information, never invents any. It is also why the reverse move is illegal: from $x \leq y$ alone you cannot recover $x < y$, because you no longer know which disjunct held.

1.5 Remark — Going Back Requires a Different Rule

To reverse the weakening you need an extra premise and a different rule:

$$\frac{\begin{array}{l} x \leq y \quad \text{i.e.} \quad (x < y) \vee (x = y) \\ x \neq y \end{array}}{\therefore x < y}$$

This is **disjunctive syllogism** — eliminate the disjunct you have ruled out. So the pair of moves fits the introduction/elimination pattern noted in Problem 4: disjunctive addition builds the $\vee$, disjunctive syllogism takes it apart. In practice this is how one upgrades a non-strict inequality: prove $x \leq y$, separately rule out $x = y$, conclude $x < y$.

1.6 Remark — Why You Would Ever Want the Weaker Statement

It can seem perverse to deliberately throw information away, but there is a very practical reason: **you must match the form a later step requires**. Many theorems are stated with $\leq$, and a theorem’s hypothesis has to be met *literally*. If a result says “if $a \leq b$ and $b \leq c$ then $a \leq c$,” you cannot feed it “ $a < b$ ” without first weakening.

The same need arises when combining inequalities of different strictness. Given $a < b$ and $b \leq c$, the cleanest route to a conclusion about a and c is often to weaken $a < b$ to $a \leq b$, apply a $\leq$ -transitivity result, and get $a \leq c$ — accepting a slightly weaker conc-

lusion in exchange for a uniform argument. This is exactly the situation Problem 4’s remark anticipated: disjunctive addition is useful precisely because it lets you *manufacture the shape* a subsequent rule demands.

1.7 Remark — The Same Move Appears Throughout Mathematics

Weakening by disjunctive addition is everywhere once you look for it:

you have	you weaken to	discarded disjunct
$x < y$	$x \leq y$	$x = y$
n is prime	n is prime or $n = 1$	$n = 1$
$A \subsetneq B$	$A \subseteq B$	$A = B$
f is strictly increasing	f is non-decreasing	f is constant somewhere

Table 1.1

In each row the second column is the *shorthand for a disjunction* containing the first, and each step is the same one-line rule. Recognising the pattern explains why so many pairs of mathematical notions come as a “strict” and a “non-strict” version.

1.8 Remark — A Caution About Inclusive Or

The definition $x \leq y \cong (x < y) \vee (x = y)$ uses the **inclusive** or of Problem 2.1.1, and it is worth being clear that nothing goes wrong even though the two disjuncts can never hold simultaneously. By **trichotomy**, exactly one of $x < y$, $x = y$, $x > y$ is true, so the disjunction always has at most one true disjunct.

Some might therefore expect an *exclusive* or. But disjunctive addition is stated for inclusive $\vee$, and using it here is still correct — inclusive or is *weaker* than exclusive or, so asserting the inclusive version is the safe move. Had $\leq$ been defined with $\oplus$, the inference $P \vdash P \oplus Q$ would in fact be **invalid** (it fails when Q is also true). The definition is

stated with $\vee$ for exactly this reason, and it is a small illustration of why the inclusive reading is the default in mathematics.